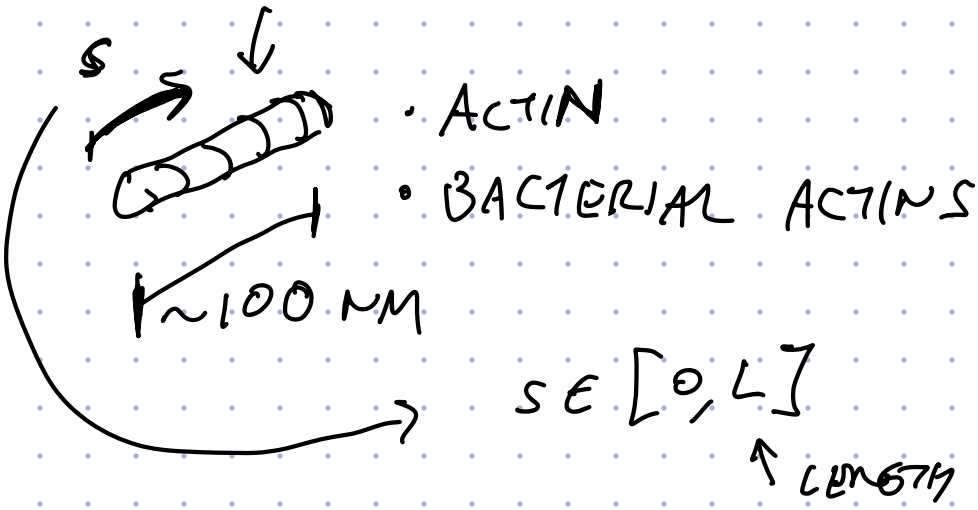
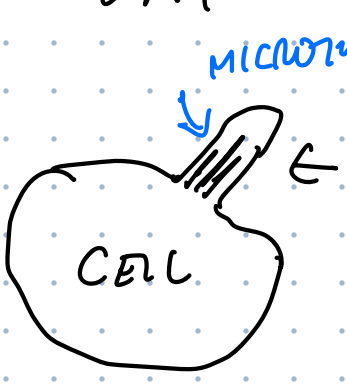


MECHANICS OF EXTENDED BODIES IN 1d, 2d, AND 3d

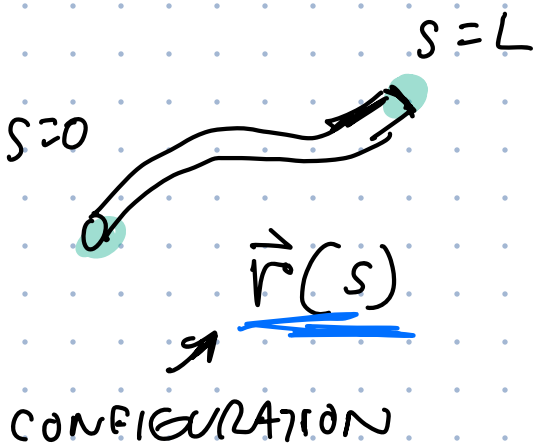
1d





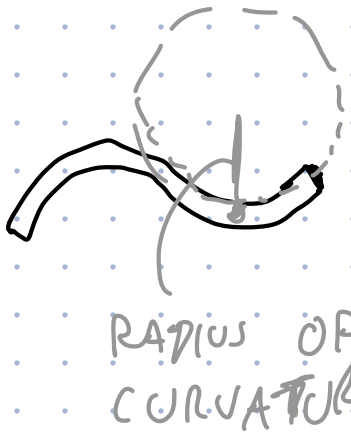


UCI:
SCOTT ATWOOD
XIAOYU SHI



$$E(\vec{r}(s)) = \int_0^L \frac{1}{2} B (\text{CURVATURE})^2 + \frac{1}{2} K (\text{STRETCH})^2 \, ds$$

$$\text{CURVATURE} = \frac{1}{\text{RADIUS OF CURVATURE}}$$



$$\left\| \frac{d}{ds} \left(\frac{d\vec{r}}{ds} \right) \right\|$$

$$\text{STRETCH} = \left\| \frac{d\vec{r}}{ds} \right\| - 1$$

- BEAM EQUATION
- KRATKY - POROD MODEL
- WORM-LIKE CHAIN

THE FILAMENT EXPERIENCES A FORCE

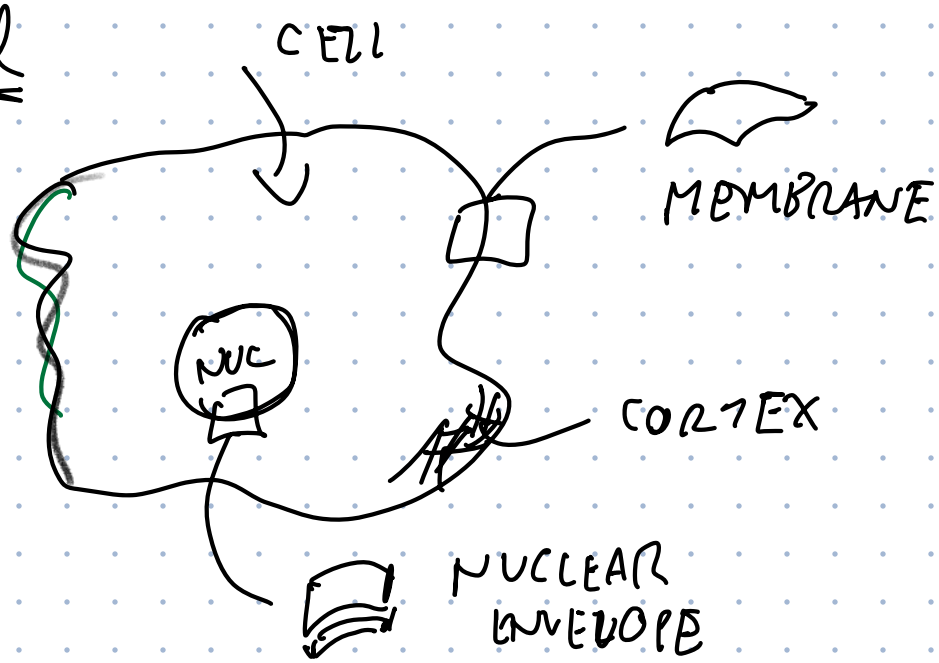
$$\vec{F} = -\frac{\delta E}{\delta \vec{r}(s)}$$

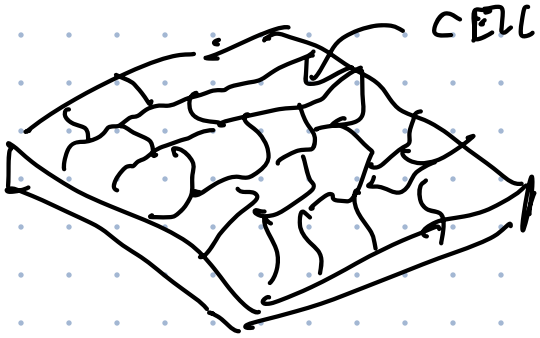
$$\vec{F} = -\frac{\partial E}{\partial \vec{r}} = -\frac{dE}{d\vec{r}}$$

EVENTUALLY, THE CONFIGURATION $\vec{r}(s)$ EVOLVES TO MINIMIZE $E(\vec{r}(s))$

ASIDE: ONE CAN USE VARIATIONAL CALCULUS, EULER-LAGRANGE EQUATIONS

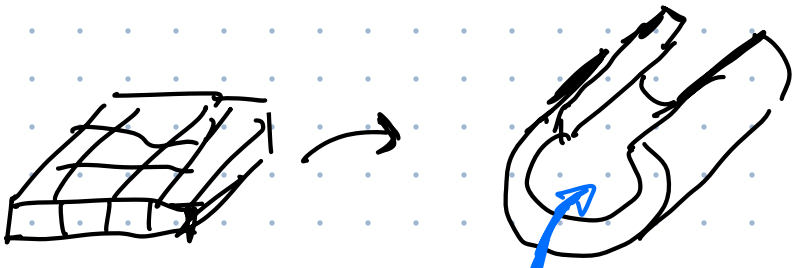
2d



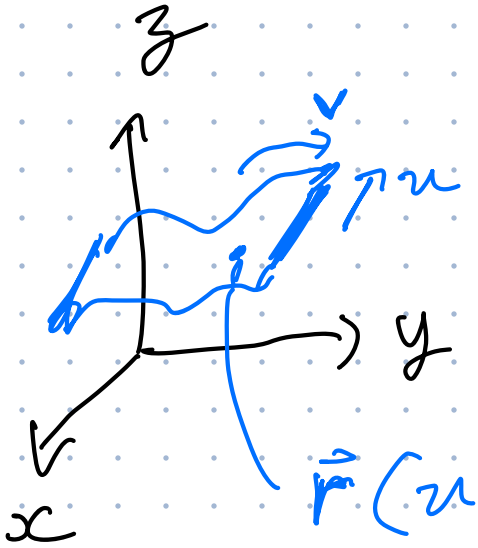


] 100 μm

EPITHELIAL SHEETS



SPINAL COLUMN



WHAT ARE THE UNITS OF B?

$$E(\vec{r}(u,v)) = \iint \frac{1}{2} B(\text{curv})^2 + \gamma \; du\,dv$$

SURFACE TENSION

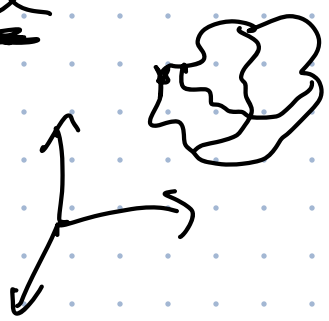
SUPPOSE CURVATURE IS CONSTANT ACROSS THE SHEET

$$E\left(\vec{r}(u,v)\right) = \left(\frac{1}{2}B\,(CURV)^2 + \gamma\right)\iint du\,dv$$

$$= \left(\frac{1}{2}B\,(CURV)^2 + \gamma\right)\cdot A$$

↑ SURFACE AREA

3d



SOLID TUMOR

$$E = \iiint \left(\phi + P\right) dV$$

↑ GENERAL ENERGY DENSITIES

↑ PRESSURE

IF ϕ AND P ARE CONSTANT ACROSS THE OBJECT, THEN

$$E = \left(\phi + P\right)\iiint dV$$

$$= \left(\phi + P\right) V$$

↑ TOTAL VOLUME